Problem 1:

$$X = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} \qquad \text{Input} \quad u(t)$$

a) $$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix}$$

$\dot{x}_1 = x_2$ (given) → for non-listed state variables and input, coefficient is 0

$\dot{x}_2 = 9.81 x_3$ (given)

$\dot{x}_3 = x_4$ (given)

$\dot{x}_4 = -2x_4 + u(t)$ (given)

All RHS are in terms of other state variables and $u(t)$, so begin to create matrix

$$\dot{X} = Ax(t) + Bu(t)$$

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 9.81 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

A B

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 9.81 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

b) Suppose that output is $x_1(t)$

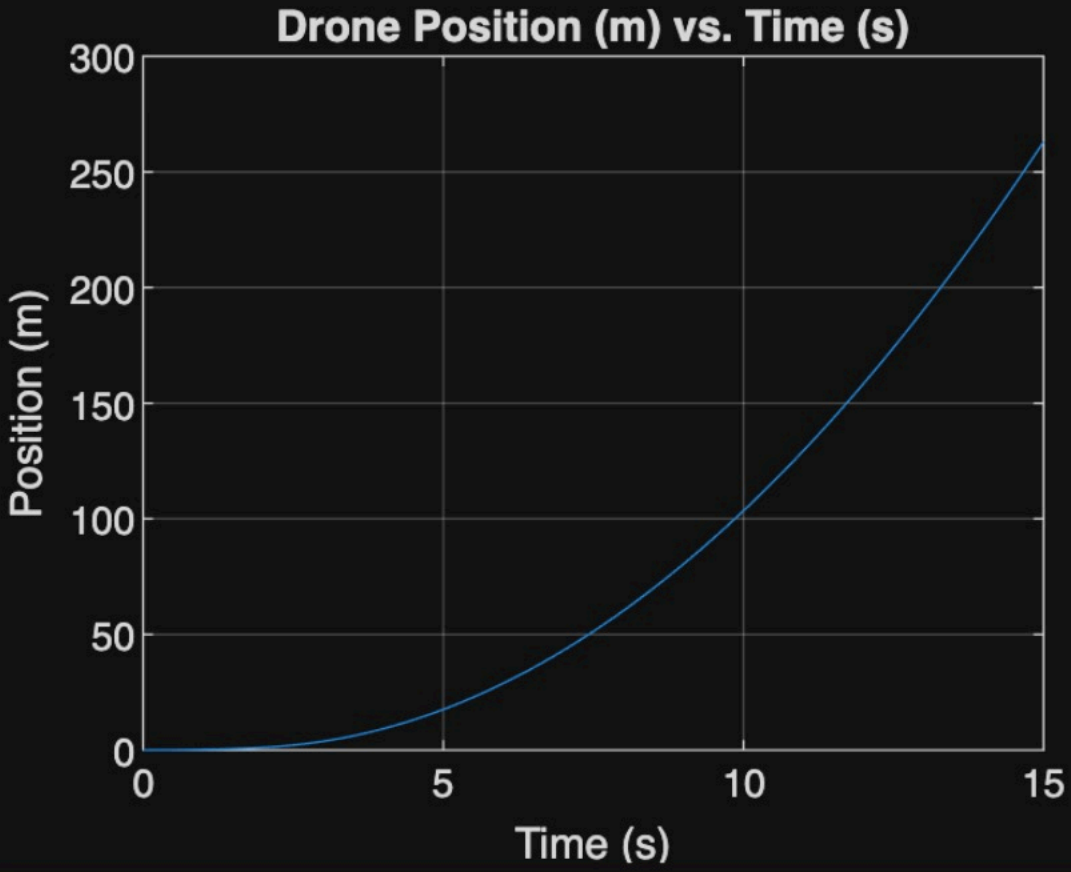
$y(t) = x_1(t)$ → all coefficients for other state variables and input is 0

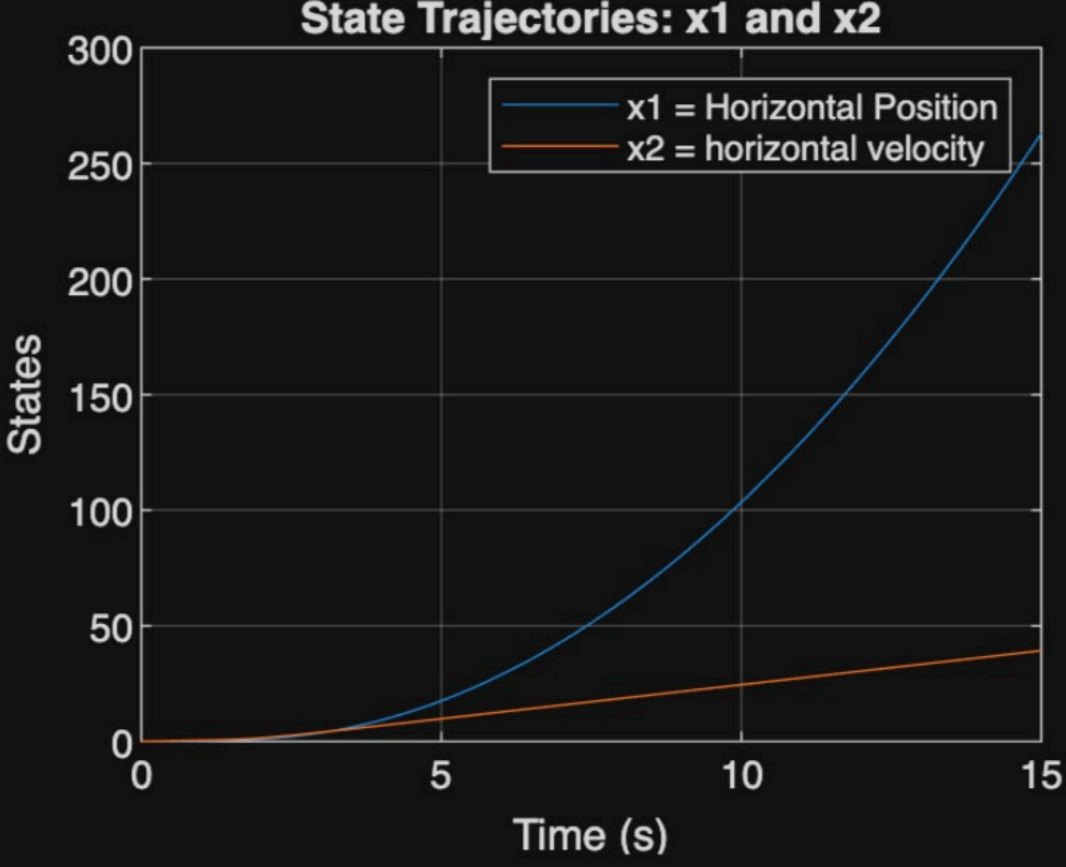
$$y = Cx(t) + Du(t)$$

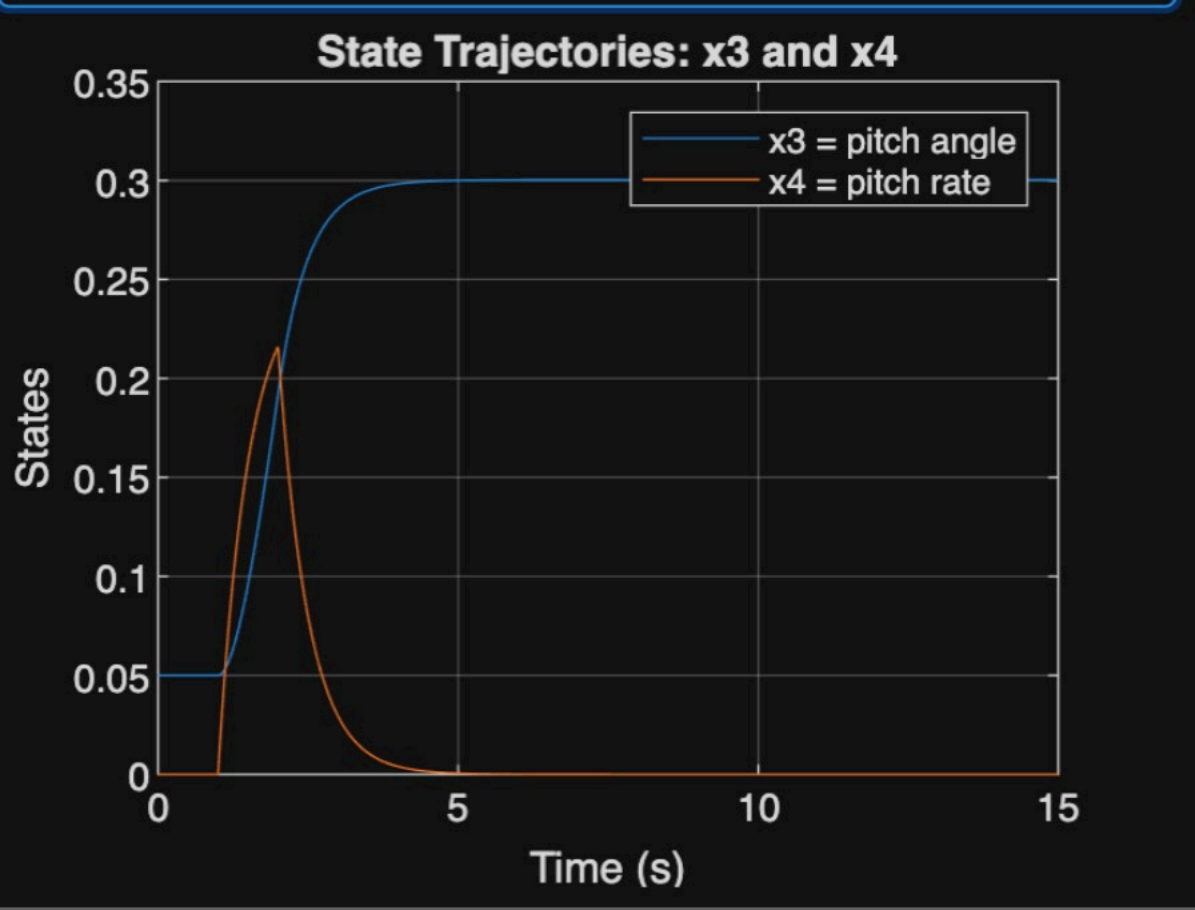
$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 0\, u(t)$$

↳ only select x_1 → position

$$\Rightarrow \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \qquad D = [0]$$


Drone Position (m) vs. Time (s)
300
250
200
150
100
50
0
Position (m)
0
5
10
15
Time (s)


State Trajectories: x1 and x2
x1 = Horizontal Position
x2 = horizontal velocity
States
Time (s)
300
250
200
150
100
50
0
0
5
10
15


State Trajectories: x3 and x4
x3 = pitch angle
x4 = pitch rate
States
Time (s)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
5
10
15

The above results make sense. x2 is rightly the derivative and rate of change of x1. x4 has a brief jump as u (input torque) is applied, which results in x3 being affected. As the torque input is released, x3 and x4 both settle.

Problem 2:

Given Equations:

$$\dot{w} = \frac{NK_m}{J_e} i - \frac{T_L}{J_e} \qquad J_e = J + N^2 J_m$$

$$\frac{di}{dt} = -\frac{R}{L} i + \frac{v}{L} - \frac{NK_m}{L} w$$

$$\dot{\theta} = w$$

output signals: θ, w

Input signal is $v(t)$

Let state vector $= x = \begin{bmatrix} \theta \\ w \\ i \end{bmatrix}$

Let input vector $= u = \begin{bmatrix} v \\ T_L \end{bmatrix}$ ↕ all functions of time, not shown for clarity

includes voltage, disturbance torque

Model:

1. $\dot{X} = Ax + Bu$

$Y = Cx + Du$

$$\dot{X} = \begin{bmatrix} \dot{\theta} \\ \dot{w} \\ \frac{di}{dt} \end{bmatrix}$$

$\dot{\theta} = w$ (given, known)

$\dot{w} = \frac{NK_m}{J_e} i - \frac{1}{J_e} T_L$ (given)

$\frac{di}{dt} = -\frac{NK_m}{L} w - \frac{R}{L} i + \frac{V}{L}$ (given)

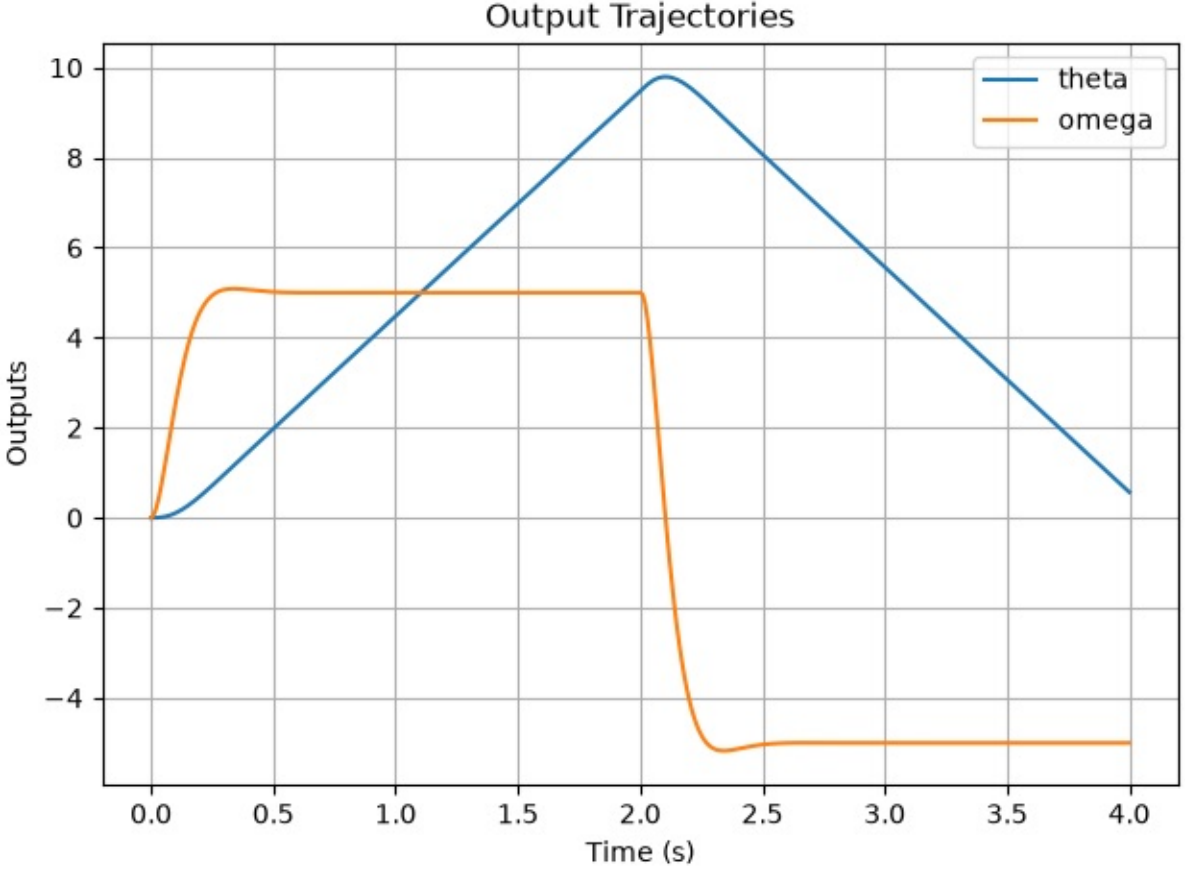
All RHS in terms of state vr, input, so modelled.

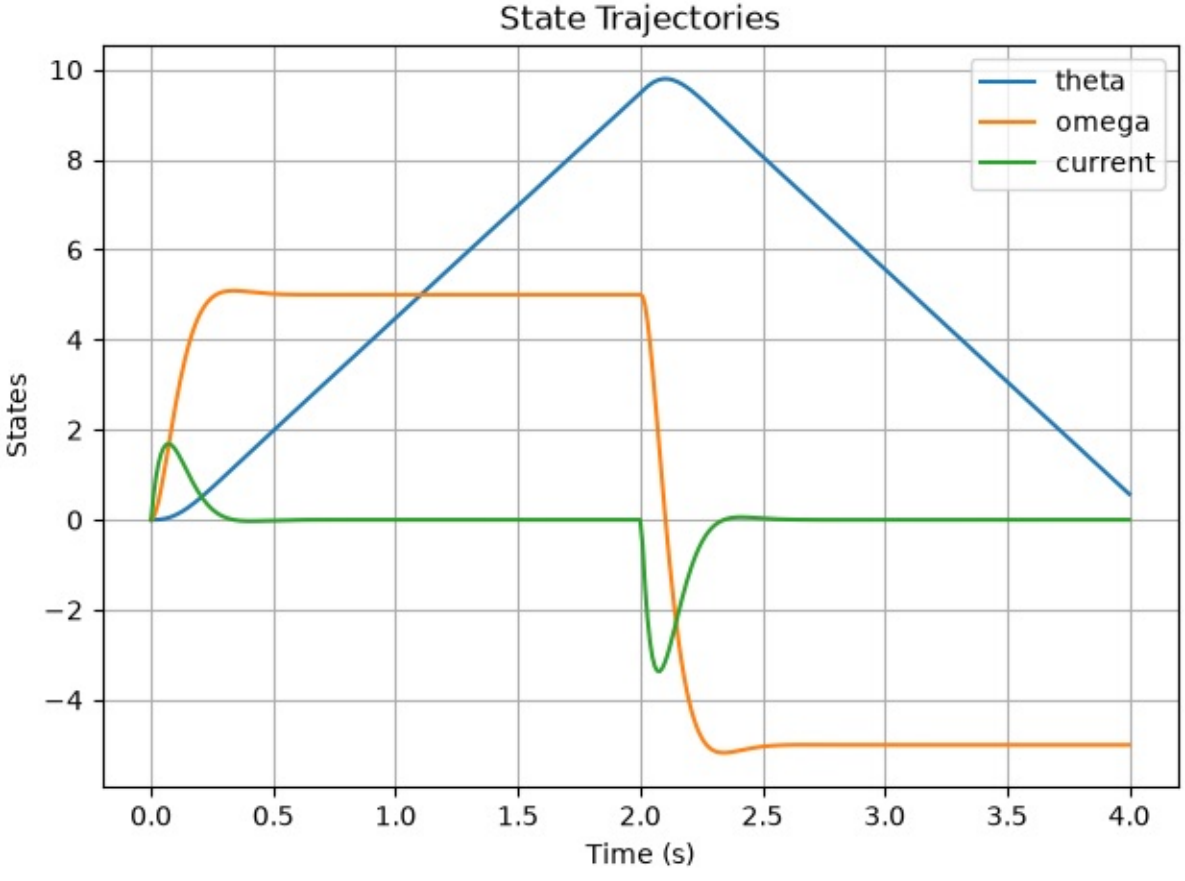
$$\dot{X} = \begin{bmatrix} \dot{\theta} \\ \dot{w} \\ \frac{di}{dt} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & \frac{NK_m}{J_e} \\ 0 & -\frac{NK_m}{L} & -\frac{R}{L} \end{bmatrix}}_{A} \begin{bmatrix} \theta \\ w \\ i \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{J_e} \\ \frac{1}{L} & 0 \end{bmatrix}}_{B} \begin{bmatrix} V \\ T_L \end{bmatrix}$$

$$y = \begin{bmatrix} \theta \\ w \end{bmatrix}$$

control inputs have no direct effect, $D = 0$ matrix

$$y = \begin{bmatrix} \theta \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{C} \begin{bmatrix} \theta \\ w \\ i \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}_{0} \begin{bmatrix} V \\ T_L \end{bmatrix}$$


Output Trajectories
theta
omega
Outputs
Time (s)
10
8
6
4
2
0
−2
−4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


State Trajectories
theta
omega
current
States
Time (s)
10
8
6
4
2
0
−2
−4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

As expected, omega is derivative of theta (as theta increases at a constant rate, omega is flat). Furthermore spikes in current represent moments where input voltage is applied (3V at 0s and -3V at 2s). Omega correspondingly changes with change in input voltage.

Problem 3:

$T_s = 0.1$

States:

$e_d(k)$ → distance error

$e_v(k)$ → velocity error

$a(k)$ → acceleration

Inputs:

$u(k)$ → control acceleration

$a_L(k)$ → acceleration of lead vehicle

1.

$$X(k) = \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix}$$

$$X(k+1) = \begin{bmatrix} e_d(k+1) \\ e_v(k+1) \\ a(k+1) \end{bmatrix}$$

put in terms of state vectors & input

$$e_d(k+1) = e_d(k) - 0.1 e_v(k) \quad \text{(given)}$$

$$e_v(k+1) = e_v(k) + 0.1 a_2(k) - 0.1 a(k) \quad \text{(given)}$$

$$a(k+1) = \beta a(k) + (1-\beta) u(k) \quad \text{(given)}$$

All RHS is in terms of state var and inputs so proceed

$$X(k+1) = \begin{bmatrix} 1 & -0.1 & 0 \\ 0 & 1 & -0.1 \\ 0 & 0 & \beta \end{bmatrix} \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0.1 \\ 1-\beta & 0 \end{bmatrix} \begin{bmatrix} u(k) \\ a_2(k) \end{bmatrix}$$

A B

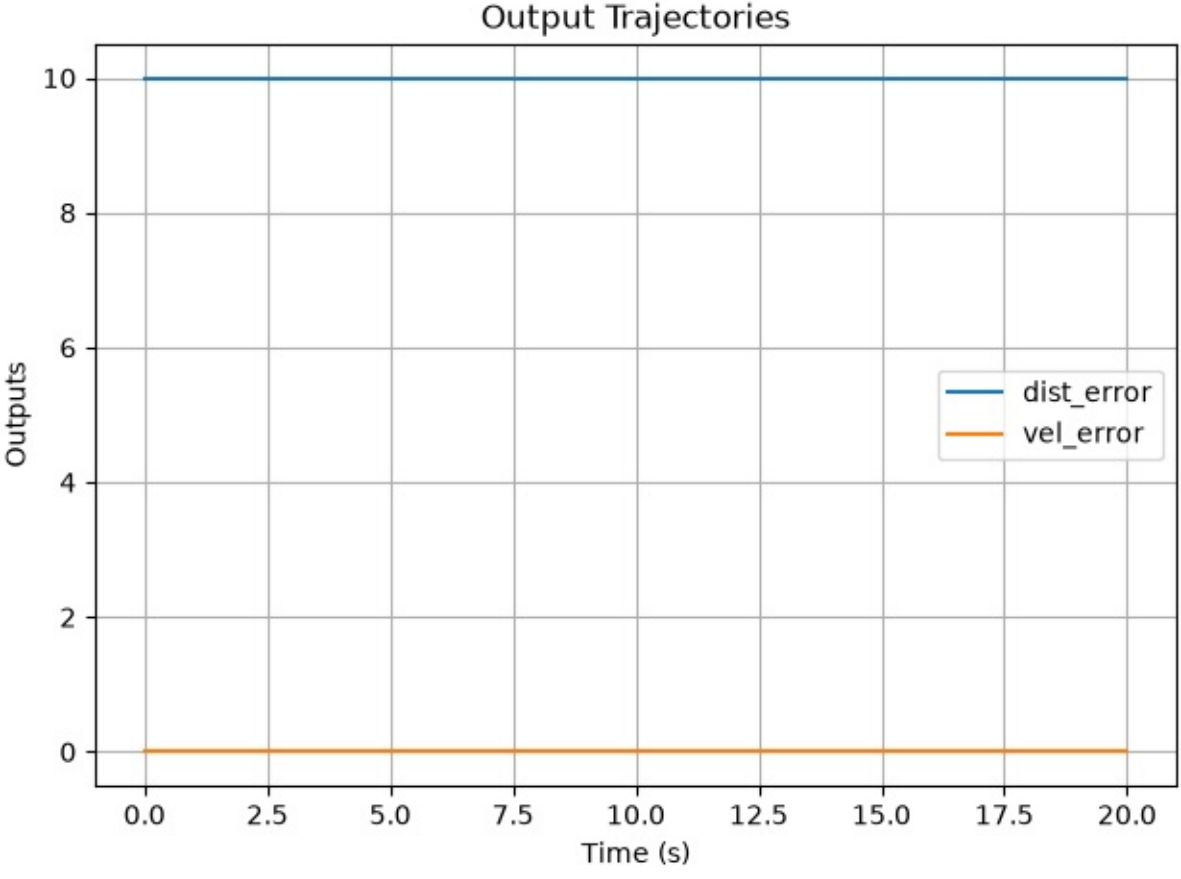
$y_1(k) = e_d(k)$

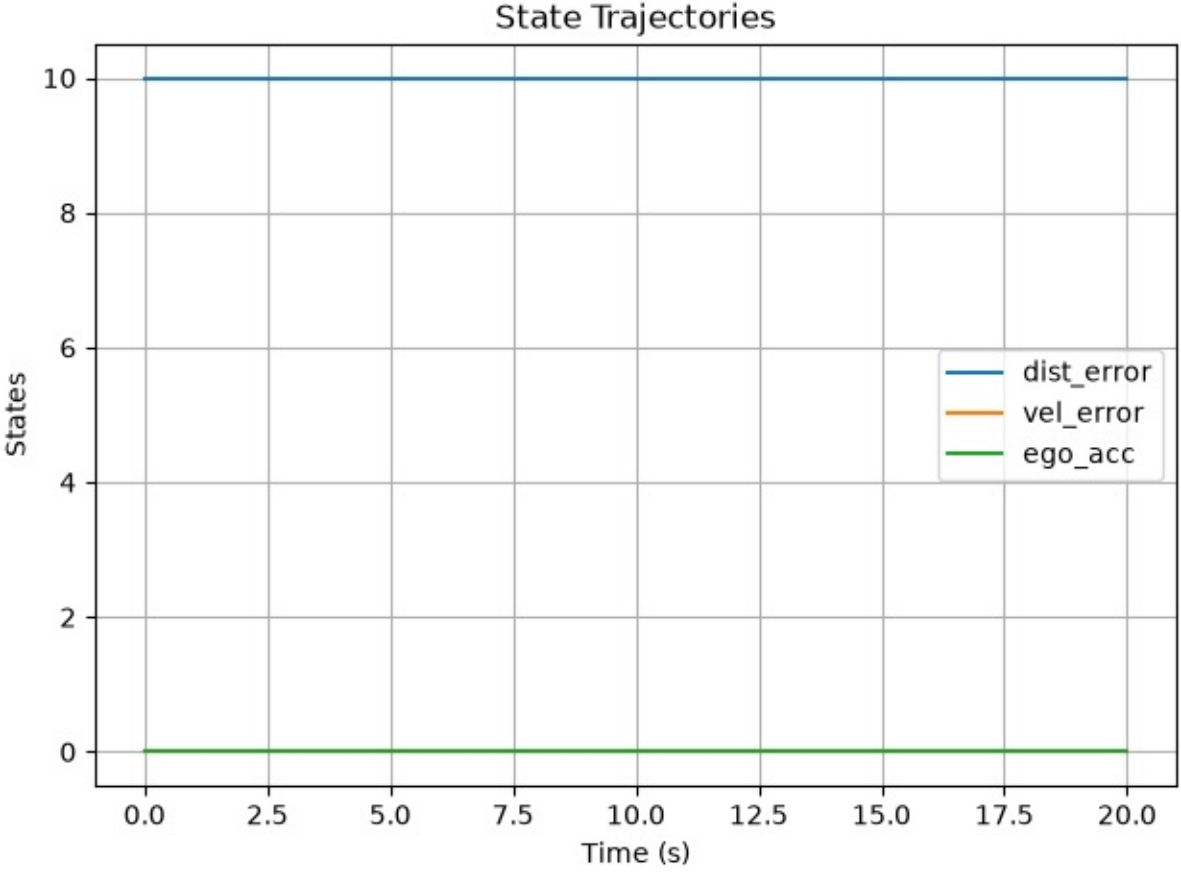
$y_2(k) = e_v(k)$

$\rightarrow$ no direct influence from control inputs $\Rightarrow D = 0$ matrix

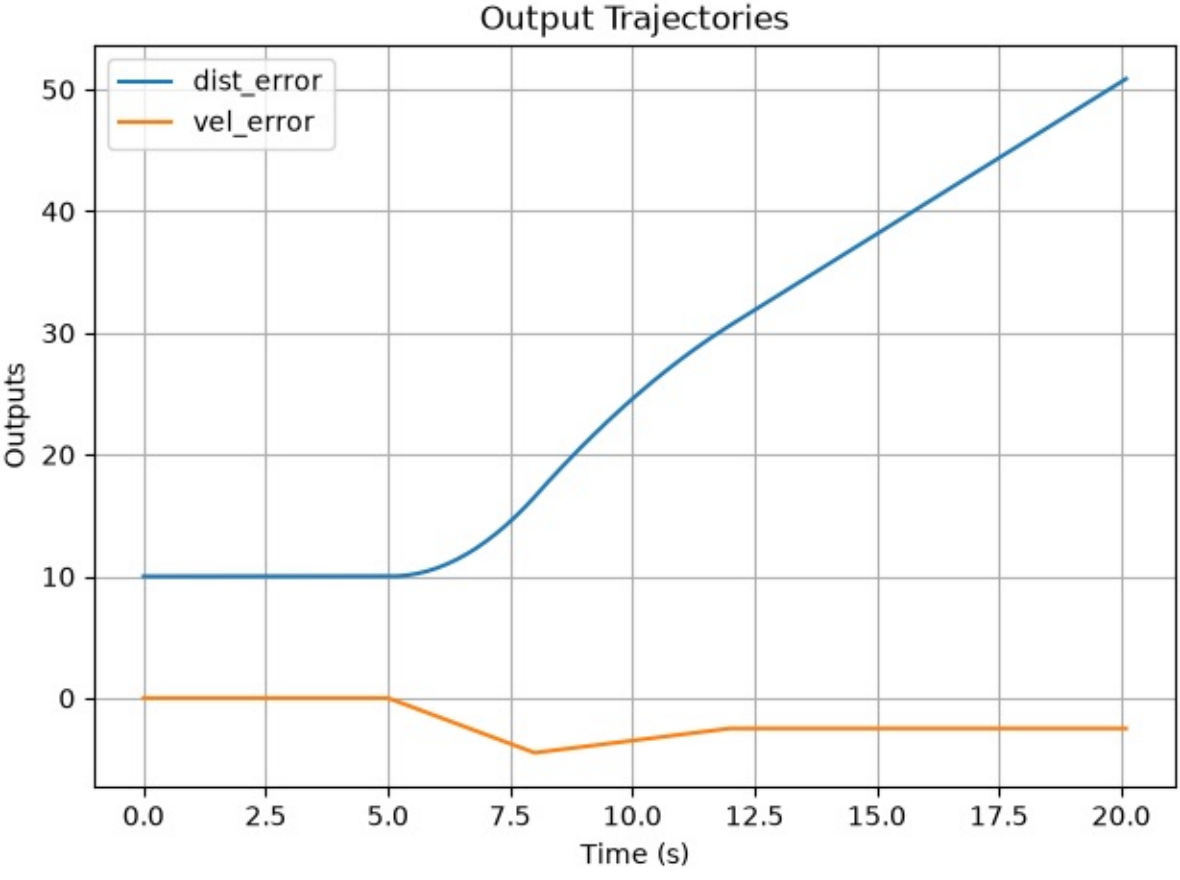
$$y(k) = \begin{bmatrix} e_d(k) \\ e_v(k) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u(k) \\ a_2(k) \end{bmatrix}$$

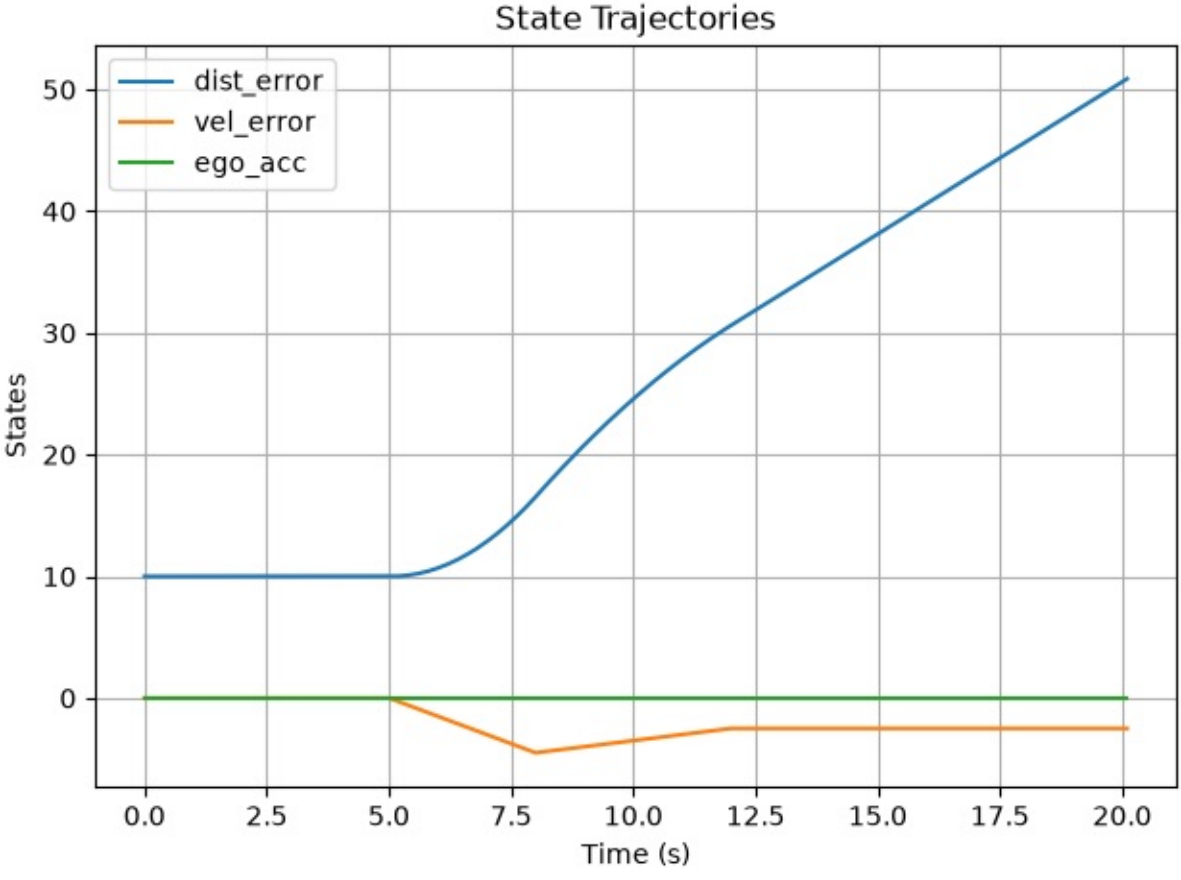
C D


Output Trajectories
Outputs
Time (s)
dist_error
vel_error
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


State Trajectories
10
8
6
4
2
0
States
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)
dist_error
vel_error
ego_acc

With distance error at 10 for all time and no perturbation,
this is the sensible output of the system. Furthermore, velocity and aceleration error should be 0 as ego car is staying constant distance of 10m away from lead for all time.


Output Trajectories
dist_error
vel_error
Outputs
50
40
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


State Trajectories
dist_error
vel_error
ego_acc
States
50
40
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

Physical Interpretation:
For all time, ego car acceleration input is 0, and initial state of ego car acceleration is 0, so it makes sense that green line in 2nd plot is flat at 0.

Between 5 and 8 seconds, the lead car is decelerating at -1.5 m/s^2. Thus, the velocity of the lead car is decreasing at a constant rate. This indicates that the relative velocity from the lead car to the ego car is decreasing constantly, as the lead car velocity is decreasing but the ego car velocity stays constant. Thus the distance error increases as the distance between the lead car and ego car shrinks.

Between 8 and 12 seconds, the opposite happens. The lead car is accelerating, so the velocity difference between the lead car and ego car increases at a constant rate. Thus, ev(k) increases at a constant rate. The distance error still increases but the distance error graph changes convexity because from 5 to 8 seconds, the lead car was decelerating but from 8 to 12 seconds the lead car is accelerating.

Finally, beyond 12 seconds, the lead car is not accelerating, so the relative velocity error stays at a constant value. However, even though the lead car's velocity is now constant, the negative value of e_v(k) indicates that the lead car is slower than the ego car. Thus, the distance error is still increasing at a constant rate as the ego car "catches up" to the lead car at a constant rate.